



FIGURE 3-24

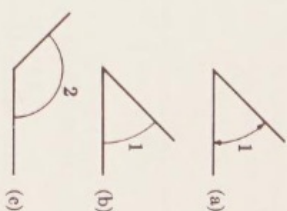


FIGURE 3-25

Remark. Note that what we call the angle is the set of points in the *sides* of the angle. Points “between” the rays are *not* part of the angle. Later we discuss the inside and the outside of an angle. It will be convenient to have a way to clearly indicate the sides of the angles. This can be done as in Fig. 3-25(a), and the angle can be numbered. Since the arrowheads indicating the sides are not vital, we can as well use the simpler notation shown in Fig. 3-25(b) and (c). *Remember that this notation is just a way of indicating the sides of the angle.*

Definition 3-7. If two lines l and m intersect at O and the four rays from O are as indicated in Fig. 3-26, then angles 1 and 2 are said to be *vertical* with respect to each other. For brevity, we say that 1 and 2 are a pair of vertical angles. Angles 3 and 4 are also vertical to each other.

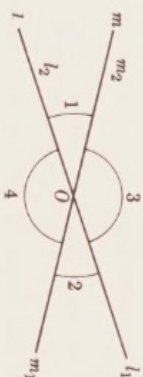


FIGURE 3-26

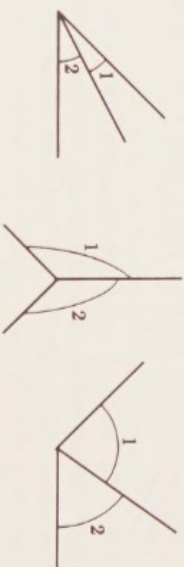


FIGURE 3-27

Definition 3-8. If two angles have a common vertex and a common side and if the other sides are on opposite sides of the line on the common side, then the angles are said to be *adjacent* to each other (Fig. 3-27).

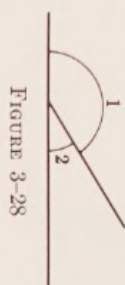


FIGURE 3-28

Definition 3-9. Two adjacent angles whose noncommon sides form a straight angle are said to be *supplementary and adjacent* to each other (Fig. 3-28).

Exercise Group 3-4

1. State in your own words the definitions of “ray,” of “angle,” and of “straight angle.”
2. How “long” are the sides of an angle?
3. If $\angle AOB$ and $\angle DOC$ are vertical to each other with B, O, D collinear, what kind of angle is $\angle AOC$?
4. If $\angle AOC$ is a straight angle and OR is a ray from O , how are angles AOR and ROC related?
5. Two lines intersect at O . With the four rays from O , how many different angles with vertex O can be found?
6. Three lines intersect at O . With the six rays from O , how many different angles with vertex O can be formed? How many pairs of vertical angles are there?
7. Draw two angles with a common vertex but which are not adjacent to each other.
8. Draw two angles with a common vertex and side but which are not adjacent.
9. State Definitions 3-7, 3-8, and 3-9 in your own words.
10. If $\angle AOB$ is not a straight angle, how many angles are there that are vertical with respect to $\angle AOB$?
11. Why is it not sensible to say, “ $\angle AOB$ is a vertical angle”?

3-6 INTERIOR AND EXTERIOR

When we draw a picture of an angle (not a straight angle) on a sheet of paper, we can tell by looking at it which points of the plane are inside and which are outside the *picture* of the angle. But in our development of geometry starting from but a few postulates, we must *define* what these sets are. (Why?) Definition 3-10 is one that can be understood by a blind person provided he knows what an angle is and that a line separates the plane. To appreciate the definition, consider Fig. 3-29. The point D lies on one side of l , which side has been shaded with vertical lines. The point C lies on one side of m , which side has been shaded with horizontal lines. The points in the plane which are shaded twice (i.e., where the vertical and horizontal lines cross) fill up the *interior* of $\angle COD$.

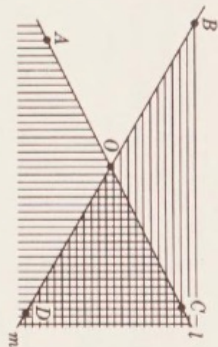


FIGURE 3-29



FIGURE 3-30

1. Copy the lines l and m and the same side of l as B and on the same side of m as C , (c) on the same side of l as D and on the same side of m as A . which are (a) on the same side of l as B and the same side of m as A , (b) on interior has been described.

Exercise Group 3-5

Definition 3-10. Consider an angle, not a straight angle, with vertex O and whose sides are the rays r_1 and r_2 on the lines l_1 and l_2 , respectively. The *interior* of the angle is the set of all points on the same side of l_1 as r_2 and on the same side of l_2 as r_1 . The *exterior* of the angle is the set of all points in the plane which are not points of the angle and are not in the interior of the angle. (Fig. 3-30)

There are several properties of the interior and exterior of angles that we can now prove. These properties are all quite obvious when we look at a figure. But even when something seems obvious from a picture, we must be sure that we have the right postulates so that our geometry agrees with our feelings about what should be true—that is, with the picture. It is really not at all obvious that our *geometry* agrees with our *picture*. Let us see that this is so by proving some “obvious” theorems. In all these theorems we shall be concerned with an angle whose vertex is O , which is not a straight angle, and whose sides are the rays r_1 and r_2 .

***Theorem 3-4.** If P and Q are both in the interior of $\angle O$, then the segment PQ is in the interior of $\angle O$.

Proof. We must show that if A is any point between P and Q , then A is in the interior of the angle (Fig. 3-31). This is easy if we use the



FIGURE 3-31

result of Exercise 16 in Exercise Group 3-2. By that exercise, since P and Q are on one side of l_1 , then A is also on that same side of l_1 . Similarly, A is on the same side of l_2 , as are both P and Q . Hence, by the definition of *interior* of an angle, A is in the interior of the angle formed by rays r_1 and r_2 .

***Theorem 3-5.** If r is a ray from O other than r_1 and r_2 , then either all the points of r other than O lie in the interior of $\angle O$ or they lie in the exterior of $\angle O$.

Proof. Let r'_1 and r'_2 be the rays from O on l_1 and l_2 other than r_1 and r_2 . (See Fig. 3-32). If $r = r'_1$ or $r = r'_2$, then r is in the exterior. If r is neither r'_1 nor r'_2 , then r lies on a definite side of l_1 and a definite side of l_2 because of the way a ray is defined. Hence r is in either the interior or exterior of $\angle O$.

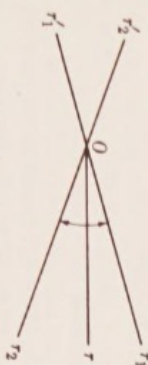


FIGURE 3-32



FIGURE 3-33

***Theorem 3-6.** If P is in the interior of $\angle O$ and Q is in the exterior of $\angle O$, then the segment PQ contains a point of the angle.

Proof. There are five possible locations of the point Q , represented by the points $Q_1, \dots, Q_5$ in Fig. 3-33. (Describe these positions in terms of sides of lines.) For positions Q_1, Q_2, Q_4, Q_5 the proofs of the theorem are quite easy and are assigned as exercises below. Hence we restrict our attention here to the position Q_3 .

Since Q_3 is on opposite sides of both l_1 and l_2 from P (why?), the segment Q_3P must contain a point R of l_1 and a point S of l_2 . We assert that either R or S is a point of the angle, that is, a point of either r_1 or r_2 . If R is on r_1 , our segment contains a point of the angle. Suppose that R is not on r_1 . Then R is on r'_1 and hence has the same relative position as Q_4 . (Why?) But then the segment RP contains a point of the angle which must be S . (Why?) Since Q_3RP and RSP , we have Q_3SP . (Why?)

Definition 3-11. The *interior* (or *inside*) of a triangle is the set of all points which are interior to each of the angles of the triangle. The *exterior* of a triangle is the set of points which are neither on the triangle nor in the interior.

If the triangle is ABC (see Fig. 3-34), an alternative definition of interior would be "the set of all points on the same side of AB as C , on the same side of BC as A , and on the same side of CA as B ." Show that these two definitions say the same thing.

We can prove the following remark by using Theorem 3-6. The remark is an "obvious" theorem that is hard to prove. We will not need this result later.



FIGURE 3-34



FIGURE 3-35

Remark. If P is in the interior of $\triangle ABC$ and Q is in the exterior, then the segment PQ contains a point of the triangle (Fig. 3-35).

Exercise Group 3-6

1. Referring to Fig. 3-36, draw an angle with vertex O . Pick a point A on one side of $\angle O$. Draw an angle with vertex A with one side the ray AO and the other side a ray intersecting the other side of $\angle O$ in B . Shade the interior of $\angle O$ and the interior of $\angle A$. Are points in both interiors also in the interior of $\triangle OBA$?

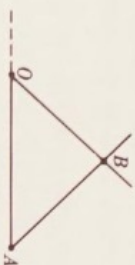


FIGURE 3-36

2. Draw a $\triangle ABC$. If a point P is on the same side of AB as C and on the same side of BC as A , show that P is not necessarily on the same side of AC as B .

3. Draw two angles whose interiors have no points in common.
4. A straight angle does not have an interior. Tell why we do not try to define interior for a straight angle.
5. State in your own words the definitions of "interior of an angle" and "interior of a triangle."
6. For a $\triangle ABC$, the point P is in the interior of $\angle A$ and not in the interior of $\angle B$. Can you be sure that P is in the exterior of $\triangle ABC$?
7. Consider the angle of Fig. 3-37, with sides l_1 and m_1 indicated on lines l and m . Tell where the point P is if:
 - (a) P is on the opposite side of m from l_1 .
 - (b) P is on the opposite side of l from m_1 .
 - (c) P is on l_1 .
 - (d) P is on m_1 .
 - (e) P is on the same side of l as m_1 .



FIGURE 3-37

- (f) P is on the same side of l as m_1 and on the same side of m as l_1 .

8. Prove that the following definition is equivalent to Definition 3-11: The interior of $\triangle ABC$ is the set of all points in the interior of $\angle A$ and also in the interior of $\angle B$. (With this definition it is not too difficult to prove the truth of the remark preceding this group of exercises.)

9. In $\angle AOB$, not a straight angle, point P is between A and B . Prove that P is in the interior of the angle.

10. In triangle ABC we have the relations AXB , AYC , and XZY . Prove that Z is in the interior of the triangle.

11. If $\angle AOB$ is not a straight angle and we have the relations XOB and XYA , prove that Y is in the exterior of $\angle AOB$.

- *12. In Theorem 3-6 suppose Q is in position Q_1 , that is, on the same side of l_2 as r_1 and on the opposite side of l_1 as r_2 (see Fig. 3-38). Show that there is



FIGURE 3-38

- *13. In Theorem 3-6 suppose Q is in position Q_2 , that is, on r'_2 (see Fig. 3-39). Show that there is a point R of r_1 between Q and P .

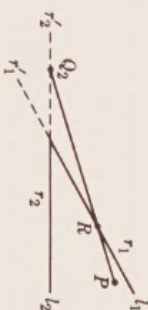


FIGURE 3-39

- *14. Tell why the positions Q_4 and Q_5 of Fig. 3-33 are much the same as positions Q_2 and Q_1 , respectively. If you have done positions Q_1 and Q_2 , why are positions Q_5 and Q_4 easy?

REVIEW OF CHAPTER 3

1. The fundamental relations of betweenness are expressed in the postulates. Study them again to be sure you know them.

2. Definitions of the following concepts were given in this chapter. Be sure that you know all of them.

- | | |
|--------------------------------------|------------------------|
| two sides of a line (see II-5) | straight angle |
| segment | vertical angles |
| triangle | adjacent angles |
| collinear (see footnote to Def. 3-3) | supplementary angles |
| two sides of a point on a line | interior of an angle |
| ray | exterior of an angle |
| angle | interior of a triangle |
| | exterior of a triangle |